Lesson plan 2026 / 27

Day & Date	Week	Period/s	Lesson Title	Subject	Chapter/Unit	Grade/ section
Week 8 · day per timetable	8	1 period (60 min)	Geometric Sequences and Series	Algebra II	Topic 6 — Exponential and Logarithmic Functions · Lesson 6-7	Grade 11 — 11B

Tools To Be Used	Resources
Quizizz (diagnostic) · Pear Deck (live practice feedback) · GeoGebra Graphing Calculator · Calculator · Canva (final product) · Whiteboards	Lesson presentation (14 slides) · "Choose Your Route" differentiation sheet · "Give it a go" classwork · "Jeddah Greening Programme, Phase 2" project task (A3) · squared paper

Competencies	Technology Embedded	Connection to student's real life
1- Identifying geometric sequences and writing recursive and explicit formulas for them. 2- Deriving the formula for a geometric series and using it, with sigma notation, to find sums.	1- Quizizz live diagnostic with instant gap analysis. 2- GeoGebra Graphing Calculator: plot the terms of a sequence as points and compare them with the exponential curve through them.	1- A greening programme that plants a fixed multiple more trees each month: how many in total, and when does the total pass a target? 2- Costing a programme in SAR from the sum of a geometric series.

Values	Soft Skills	Hard Skills
1- Precision — naming the first term, ratio and number of terms before using a formula. 2- Sustainability — planning resources for national targets.	1- Collaboration in assigned group roles. 2- Explaining each step as following from the line before it.	1- Writing recursive and explicit formulas for geometric sequences. 2- Using the geometric series formula and sigma notation to find sums.

Lesson Integral Parts			
Objectives / skills	**Teacher Strategies & Student Activities**	**Assessment**	**Time**
1- Identify geometric sequences. 2- Write recursive and explicit formulas for geometric sequences. 3- Identify geometric series. 4- Derive the formula for geometric series. 5- Use the formula for geometric series to find sums. *Essential question: How can we use geometric sequences to model real-world situations?* *Vocabulary: 10 terms — listed in the FIKR plan.*	FIKR Phase 1 — Diagnose. 3-question Quizizz on a power before a product, the shape of the sum formula, and adding versus multiplying. Results read as a gap map; students routed to re-teach, the examples, or Investigate. FIKR Phase 2 — Targeted Instruction. Re-teach powers for flagged students. Whole class: identify a geometric sequence; recursive and explicit formulas; geometric series, sigma notation and the derivation of the sum formula; using the formula. FIKR Phase 3 — Practice. Two "Give it a go" problems worked together; students then choose a route and may switch at any time. Pear Deck feedback; 2-minute micro-conferences. FIKR Phase 4 — Production. The Jeddah greening programme: both formulas, a six-month total and its cost in SAR, and the first month the total passes 10 000 trees. AI as critic only. FIKR Phase 5 — Proof of Learning. "Time to Check": find a ratio and the next term, a recursive rule and its sixth term, and a sum in sigma notation. No notes, partner or AI. FIKR Phase 6 — Smart Production. Students refine their reasoning into a final product with one worked non-example, and publish it.	Diagnostic quiz — gap map, not a grade. Quick check for understanding on whiteboards (80% threshold). "Give it a go" questions — self-marked, teacher circulates. Project task sheet against its "Done when…" criteria. "Time to Check" — Mastery Gate, scored live. PASS → Enrichment & Challenge. NOT YET → Targeted Learning Clinic next lesson. Final product graded on reasoning and clarity, not correctness alone.	6 min 18 min 12 min 12 min 6 min 6 min **Total 60 min**

Lesson Closure:	Peer show-and-tell (2 min), then the reflection: "How do you tell an arithmetic sequence from a geometric one?"
Reflection for Improvement:	
Homework:	Upload the final product to the LMS portfolio; Pear Deck practice set; flipped video for Lesson 9-1.
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